1.22 26/07/17 - The Rank Formula, Examples and Applications, Mazur's Theorem and Rank Records

We have at this point shown that if $E/\mathbb{Q}$ is an elliptic curve, then $E(\mathbb{Q})$ is a finitely generated abelian group (1.15.8) and hence (1.21.4) that $E(\mathbb{Q}) \cong_{\mathsf{Ab}} \mathbb{Z}^r \oplus \operatorname{Tors} E(\mathbb{Q})$ for a unique integer $r \in \mathbb{Z}_{\geq 0}$ (depending on E) and finite group $\operatorname{Tors} E(\mathbb{Q})$. This integer r is called the *(algebraic) rank* of the elliptic curve E, and is written as $r = \operatorname{rank} E$. This rank encodes important arithmetic information about rational points on E; for instance, E has finitely many rational points iff $\operatorname{rank} E = 0$. The Nagell-Lutz Theorem (1.13.7) gives an algorithmic procedure to compute $\operatorname{Tors} E(\mathbb{Q})$, but how do we actually compute the rank of a given elliptic curve E over $\mathbb{Q}$?

Remark 1.22.1. If we look at the proofs of 1.18.1, 1.19.1, and 1.21.1 more carefully, we will note that (up to a suitable analog of 1.14.8), we have basically proven the weak Mordell-Weil theorem (1.15.9) for not just $\mathbb{Q}$ but any number field $K/\mathbb{Q}$. The proof of the full Mordell-Weil Theorem (1.8.1, the analog of 1.15.8 for number fields K) is then just one step away: namely, developing the theory of heights as in §1.16 over a general number field, for then we can invoke 1.15.11. This is done systematically, for instance, in [8, Chapter VIII].

1.22.A The Rank Formula

As in §1.20, we will henceforth assume that the elliptic curve E has a nontrivial rational 2-torsion point (but see 1.22.5(b)). This means that E can be brought into the form $E = E_{a,b} : y^2 = x(x^2 + ax + b)$ for some integers $a, b \in \mathbb{Z}$ with $b(a^2 - 4b) \neq 0$. As in 1.20, we can then let $(a', b') := (-2a, a^2 - 4b)$, and let $E' : (y')^2 = (x')^3 + a'(x')^2 + b'x'$. Then 1.20.13 gives us morphisms $\phi : E \to E'$ and $\phi' : E' \to E$ with the property that $\phi' \circ \phi = [2]_E$ and $\phi \circ \phi' = [2]_{E'}$. Finally, 1.21.1 gives us a map $\alpha : E(\mathbb{Q}) \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ such that $\ker \alpha = \phi'(E'(\mathbb{Q}))$ and $\operatorname{im} \alpha \subset \mathbb{Q}(S, 2)$, where $S = \{p : p \mid b\}$; then $\operatorname{coker} \phi'(\mathbb{Q}) \xrightarrow{\sim} \operatorname{im} \alpha$. Similarly, we have a map $\alpha' : E'(\mathbb{Q}) \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ with the property that $\ker \alpha' = \phi(E(\mathbb{Q}))$ and $\operatorname{im} \alpha' \subset \mathbb{Q}(S', 2)$, with $S' := \{p : p \mid b'\}$; again $\operatorname{coker} \phi(\mathbb{Q}) \xrightarrow{\sim} \operatorname{im} \alpha'$.

Now we can consider the sequence of group homomorphisms

$$E(\mathbb{Q}) \xrightarrow{\phi(\mathbb{Q})} E'(\mathbb{Q}) \xrightarrow{\phi'(\mathbb{Q})} E(\mathbb{Q})$$

which compose to the duplication map $[2] : E(\mathbb{Q}) \to E(\mathbb{Q})$. We can apply 1.20.3(b) to this to obtain the exact sequence

$$0 \to \ker \phi(\mathbb{Q}) \to E[2](\mathbb{Q}) \to \ker \phi'(\mathbb{Q}) \to \operatorname{coker} \phi(\mathbb{Q}) \to E(\mathbb{Q})/2E(\mathbb{Q}) \to \operatorname{coker} \phi'(\mathbb{Q}) \to 0.$$

We know that $\ker \phi(\mathbb{Q}) = \{\mathscr{O}, T\}$ and $\ker \phi'(\mathbb{Q}) = \{\mathscr{O}', T'\}$. Further, we know that $\operatorname{coker} \phi(\mathbb{Q}) \xrightarrow{\sim} \operatorname{im} \alpha$ and $\operatorname{coker} \phi'(\mathbb{Q}) \xrightarrow{\sim} \operatorname{im} \alpha'$. Using these facts along with 2.5.11(d) tells us immediately that

$$2 \cdot 2 \cdot \# E(\mathbb{Q})/2E(\mathbb{Q}) = \# E[2](\mathbb{Q}) \cdot \# \operatorname{im} \alpha \cdot \# \operatorname{im} \alpha',$$

which can be rearranged to be

$$\frac{\# E(\mathbb{Q})/2E(\mathbb{Q})}{\# E[2](\mathbb{Q})} = \frac{\# \operatorname{im} \alpha \cdot \# \operatorname{im} \alpha'}{4}. \tag{1.22.2}$$

Now 1.21.4 tells us that $E(\mathbb{Q}) \cong_{\mathsf{Ab}} \mathbb{Z}^r \oplus T$ for some unique integer $r \in \mathbb{Z}_{\geq 0}$ (namely $r = \operatorname{rank} E$) and some unique finite group T (up to isomorphism). In this case, we clearly have that

$$E(\mathbb{Q})/2E(\mathbb{Q}) \cong_{\mathsf{Ab}} (\mathbb{Z}/2\mathbb{Z})^r \oplus (T/2T) \text{ and } E[2](\mathbb{Q}) \cong T[2].$$

In particular,

$$\# E(\mathbb{Q})/2E(\mathbb{Q}) = 2^r \cdot \# T/2T \text{ and } \# E[2](\mathbb{Q}) = \# T[2].$$

However, it is not hard to see at all that $\# T/2T = \# T[2]$ (you could use 1.21.4, or apply 2.5.11(c) cleverly!), and so this is enough to conclude that

$$2^r = \frac{\# \operatorname{im} \alpha \cdot \# \operatorname{im} \alpha'}{4}. \tag{1.22.3}$$

The formula on the right side is completely symmetric in E and E'. We have deduced

Theorem 1.22.4 (The Rank Formula). In the above setup, we have that

$$\operatorname{rank} E = \operatorname{rank} E' = \log_2 \# \operatorname{im} \alpha + \log_2 \# \operatorname{im} \alpha' - 2.$$

Given our characterization 1.21.2 of the images of α and α' in terms of homogeneous spaces, we have a strategy for determining the ranks of elliptic curves (but see 1.22.16), which we will use in the next section to compute lots of examples.

Remark 1.22.5.

(a) The result of 1.22.4 implies that the two-isogenous curves E and E' have the same rank over $\mathbb{Q}$. It can be shown in general that if $E/\mathbb{Q}$ and $E'/\mathbb{Q}$ are two elliptic curves are isogenous over $\mathbb{Q}$ (1.20.4 and 1.20.7(b)), then $\operatorname{rank} E = \operatorname{rank} E'$. Indeed, the same is true of elliptic curves isogenous over any number field (1.8.1); this follows immediately from the fact that the kernel of a nonconstant isogney is finite and that being isogenous is an equivalence relation (1.20.7(b)).

(b) When the elliptic curve $E/\mathbb{Q}$ does not have a nontrivial rational two-torsion point, then the above strategy cannot be used to compute $\operatorname{rank} E$. This can be done either by the use of a little algebraic number theory (2.5.17), or by using the theory of binary quartics; this is explained, for instance, in Cremona's book ([17, Chapter III]).

1.22.B Examples and Applications

Let's now work through some examples.

Example 1.22.6. Let $E : y^2 = x^3 - x$, so that $(a, b, a', b') = (0, -1, 0, 4)$. In this case, $\operatorname{im} \alpha \subset \mathbb{Q}(\emptyset, 2) = \{[\pm 1]\}$, but the image certainly contains $[1] = \alpha(\mathscr{O})$ and $[-1] = \alpha(T)$. This implies that $\operatorname{im} \alpha = \{\pm 1\}$, and hence $\# \operatorname{im} \alpha = 2$.

The corresponding two-isogenous curve is $E' : y^2 = x^3 + 4x$. Here $\operatorname{im} \alpha' \subset \mathbb{Q}(\{2\}, 2) = \{[\pm 1], [\pm 2]\}$. The image again certainly contains [1]. By 1.21.2, we have that $[-1] \in \operatorname{im} \alpha'$ iff there is a nontrivial solution to $w^2 = -u^4 - 4v^4$, which visibly does not exist (there is an obstruction at infinity!). The same reasoning implies that $[-2] \notin \operatorname{im} \alpha'$. Finally, we have that $[2] \in \operatorname{im} \alpha$ iff there is a nontrivial solution to the equation $w^2 = 2u^4 + 2v^4$, which does in fact exist–we could take $(u, v, w) = (1, 1, 2)$. This proves that $\operatorname{im} \alpha' = \{[1], [2]\}$, and hence $\# \operatorname{im} \alpha' = 2$.

At this point, we are done and can use 1.22.4 to conclude that $\operatorname{rank} E = \operatorname{rank} E' = 0$. Since $\operatorname{rank} E = 0$, we conclude that $E(\mathbb{Q}) = \operatorname{Tors} E(\mathbb{Q})$, which we find using Nagell-Lutz (1.13.7) to be $\{\mathscr{O}, (0, 0), (\pm 1, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/2$. Therefore, we have determined all rational points of E: we have that

$$E(\mathbb{Q}) = \{\mathscr{O}, (0, 0), (\pm 1, 0)\}.$$

Similarly, $\operatorname{rank} E' = 0$, so $E'(\mathbb{Q}) = \operatorname{Tors} E'(\mathbb{Q})$, which again by Nagell-Lutz (1.13.7) is easily seen to be $\{\mathscr{O}, (0, 0), (2, \pm 4)\} \cong_{\mathsf{Ab}} \mathbb{Z}/4$. Therefore, we have determined all the rational points of E' too: we have that

$$E'(\mathbb{Q}) = \{\mathscr{O}, (0, 0), (2, \pm 4)\}.$$

Let's now derive some delicious consequences.

Corollary 1.22.7. There does not exist a rational right-angled triangle with area a perfect square.

Proof. Suppose there exists such a right angled triangle. We can rescale it to have coprime integer sides; then 1.2.3 tells us that it must have sides of the form $(u^2 - v^2, 2uv, u^2 + v^2)$ for some integers $u > v > 0$. The area of this triangle is exactly $uv(u^2 - v^2)$; therefore, if this area is a square, then there is an integer $w > 0$ such that $w^2 = uv(u^2 - v^2)$. If we divide throughout by v^4, then we see immediately from this equation that $(u/v, w/v^2) \in E(\mathbb{Q})$, where $E : y^2 = x^3 - x$ as in 1.22.6. However, we saw in that example that E does not have *any* rational points with nonzero finite y-coordinate, and this is the required contradiction. ∎

You will investigate some more questions relating to the areas of rational right-angled triangles on the last Exercise Sheet. We now use the theory of elliptic curves to give a second proof of Fermat's Last Theorem for $n = 4$.

Corollary 1.22.8. Let $X, Y, Z \in \mathbb{Z}_{\geq 0}$ be such that $X^4 + Y^4 = Z^4$. Then $XYZ = 0$.

Proof. If $Z = X$, then $Y = 0$. Else, 2.1.12(b) shows (spoiler alert!) that the point

$$(x, y) := \left(2\left(\frac{Z+X}{Z-X}\right), \frac{4Y^2}{(Z-X)^2}\right)$$

is a rational point other than $\mathcal{O}$ on the curve $E' : y^2 = x^3 + 4x$ of 1.22.6. By our complete understanding of $E'(\mathbb{Q})$ in 1.22.6, either $Y = 0$, or

$$2\left(\frac{Z+X}{Z-X}\right) = 2,$$

which implies that $X = 0$. ∎

You will use a different curve to give a second proof of 1.3.1 using elliptic curve theory on the last Exercise Sheet.

Remark 1.22.9. We have shown in 1.3.1 that there does not exist an integer square which is the sum of two integral fourth powers. As a natural next question, it makes sense to relax the condition of being square to something weaker. One such weakening is the notion of being *powerful*: an integer $n \in \mathbb{Z}$ is said to be powerful iff for each prime p, if $p \mid n$, then $v_p(n) \geq 2$. Can a powerful integer be the sum of two integral fourth powers? Yes: for instance, $32 = 16 + 16$. This is silly. A better question to ask is: can a powerful integer be the sum of two *coprime* integral fourth powers? This is a much more subtle question. The answer to this particular question turns out to be "yes", but the smallest integer so expressible is

$$\begin{aligned} &3088257489493360278725196965477359217 \\ =& 427511122^4 + 1322049209^4 \\ =& 17^3 \cdot 73993169^2 \cdot 338837713^2. \end{aligned}$$

This is a *huge* number ($\approx 3.08 \times 10^{36}$), and it was not found–nor this result proven–using a brute-force search. This number arises rather from a suitable rational point on a quadratic twist (namely $\mathcal{E}_{17} : y^2 = x^3 - 68^2 x$) of our E from 1.22.6. For a detailed study of this particular problem, we refer the reader to [16].

Let's now look at another example.

Example 1.22.10. Let $E : y^2 = x^3 + 5x^2 + 4x$, so that $(a, b, a', b') = (5, 4, -10, 9)$. In this case, $\operatorname{im}\alpha \subset \mathbb{Q}(\{2\}, 2) = \{[\pm 1], [\pm 2]\}$. As above, $\operatorname{im}\alpha \ni [1]$. By 1.21.2, for each squarefree divisor b_1 of $b = 4$, to determine whether each of $[b_1] \in \operatorname{im}\alpha$, we need to study whether a homogeneous space has a nontrivial rational point; let's do this for $b_1 = -1, 2, -2$ below.

(a) If $b_1 = -1$, we are looking at the equation $w^2 = -u^4 + 5u^2v^2 - 4v^4$, which does have a nontrivial rational solution $(1, 1, 0)$. This proves that $[-1] \in \operatorname{im}\alpha$.
(b) If $b_1 = 2$, we are looking at $w^2 = 2u^4 + 5u^2v^2 + 2v^4$, which also has a nontrivial rational solution $(1, 1, 3)$. This proves that $[2] \in \operatorname{im}\alpha$ as well.
(c) To determine if $b_1 = -2$ works, we could look at the relevant homogeneous space, or we could argue that $\operatorname{im}\alpha$ is a subgroup of $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so if it contains both $[-1]$ (thanks to (a)) and $[2]$ (thanks to (b)), then it also contains $[-1] \cdot [2] = [-2]$.

In particular, we conclude that $\operatorname{im}\alpha = \{[\pm 1], [\pm 2]\}$, so that $\#\operatorname{im}\alpha = 4$.

The relevant two-isogenous curve is $E' : y^2 = x^3 - 10x^2 + 9x$. Now, $\operatorname{im}\alpha' \subset \mathbb{Q}(\{3\}, 2) = \{[\pm 1], [\pm 3]\}$. As above, let's use 1.21.2 and work through the cases:

(a) We have that $[-1] \in \operatorname{im}\alpha$ iff $w^2 = -u^4 - 10u^2v^2 - 9v^4$ has a nontrivial rational solution, but it doesn't–this equation has no nontrivial solution over $\mathbb{R}$ (i.e., is obstructed at infinity).
(b) We can exclude $[-3]$ for the same reason as in (a).

(c) Finally, $[3] \in \operatorname{im}\alpha$ iff there is a nontrivial rational solution to $w^2 = 3u^4 - 10u^2v^2 + 3v^4$. This equation is more subtle. It clearly has solutions over $\mathbb{R}$, so our technique for (a) and (b) above is not going to work. Let's try to argue differently. Suppose that (u, v, w) is a solution, and we clear out denominators so that $u, v, w \in \mathbb{Z}$. Then reducing the equation mod 3 tells us that

$$w^2 \equiv -u^2v^2 \pmod 3,$$

which implies that $3 \mid w$ and $3 \mid uv$. Write $w = 3w_1$ for some $w_1 \in \mathbb{Z}$. Since $3 \mid uv$, we have that either $3 \mid u$ or $3 \mid v$; since u and v play symmetric roles, we might as well assume that $3 \mid u$ and write $u = 3u_1$ for some $u_1 \in \mathbb{Z}$. The above equation can then be rewritten as

$$3w_1^2 = 81u_1^4 - 30u_1^2v^2 + v^4.$$

This equation implies now that $3 \mid v$, and a quick inspection then shows that $3 \mid w_1$ as well. At this point, we have shown that if (u, v, w) is an integer solution to $w^2 = 3u^4 - 10u^2v^2 + 3v^4$, then so is $(u/3, v/3, w/9)$. Infinite descent shows that this equation is therefore *not* solvable over $\mathbb{Q}$ (it is even obstructed at 3, or equivalently has no solutions over $\mathbb{Q}_3$); in particular, $[3] \notin \operatorname{im}\alpha'$.

We have now shown that $\operatorname{im}\alpha = \{[1]\}$, and so $\#\operatorname{im}\alpha' = 1$.

At this point, we are done and can use 1.22.4 to conclude that $\operatorname{rank} E = \operatorname{rank} E' = 0$. Since $\operatorname{rank} E = 0$, we conclude that $E(\mathbb{Q}) = \operatorname{Tors} E(\mathbb{Q})$, which we can find using Nagell-Lutz (1.13.7) to be

$$E(\mathbb{Q}) = \operatorname{Tors} E(\mathbb{Q}) = \{\mathscr{O}, (0, 0), (-1, 0), (-4, 0), (2, \pm 6), (-2, \pm 2)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/4.$$

(The points can be found with Nagell-Lutz; the group structure can be found by a little experimentation–left as an exercise!). Similarly, all points on E' can be found by using the Nagell-Lutz Theorem.

Exercise 1.22.11. Check that the the description of the group structure of $E(\mathbb{Q}) = \operatorname{Tors} E(\mathbb{Q})$ in 1.22.10 is correct. Next, use the Nagell-Lutz Theorem (1.13.7) to find all rational points on E', and determine the group structure of $E'(\mathbb{Q})$.

Remark 1.22.12. We could have deduced the result of 1.22.10 with less computation. Indeed, it follows already from $\#\operatorname{im}\alpha' = 1$, the fact that $b = 4$, the fact that the rank is a nonnegative integer, and 1.22.4 that $\operatorname{rank} E = \operatorname{rank} E' = 0$. This illustrates the general principle that the rank of an elliptic curve E of the form $y^2 = x^3 + ax^2 + b$ as above is bounded above by some quantity depending only on the number of primes dividing b and $b' = a^2 - 4b$; this is an exercise on the next sheet.

As another delicious corollary, we can now derive the following result stated by Fermat in 1640 and first proven by Euler in 1780.

Corollary 1.22.13 (Squares in Arithmetic Progression)**.** There is no nonconstant four-term sequence of integer squares in arithmetic progression.

Proof. This is a fun consequence of the fact that for $E : y^2 = x^3 + 5x^2 + 4x$ as in 1.22.10 as above, we have $\operatorname{rank} E = 0$ (see 2.3.5). ■

You will be asked to investigate a similar result about cubes and fourth powers in arithmetic progression on the last exercise sheet.

Let's work through one final (extended) example together.

Example 1.22.14. Let p be a prime, and let $E_p : y^2 = x^3 + px$ so that $(a, b, a', b) = (0, p, 0, -4p)$. We already have enough information to conclude that $\operatorname{rank} E_p \leq 2$ for odd p, and that $\operatorname{rank} E_2 \leq 1$ (check!).

We know first that $\operatorname{im}\alpha \subset \mathbb{Q}(\{p\}, 2) = \{[\pm 1], [\pm p]\}$. As above, we know already that $[1], [p] \in \operatorname{im}\alpha$. At this point, it only remains to determine whether $b_1 = -1$ works (check!), for which by 1.21.2 we are trying to solve $w^2 = -u^4 - pv^4$. This equation is evidently obstructed at infinity, and so we conclude that $\operatorname{im}\alpha = \{[1], [p]\}$; in particular, $\#\operatorname{im}\alpha = 2$.

The corresponding two-isogenous curve is $E'_p : y^2 = x^3 - 4px$, and here $\operatorname{im}\alpha' \subset \mathbb{Q}(\{2, p\}, 2)$. Here the analysis of the case $p = 2$ diverges from that of the case $p > 2$; I will leave the easier case of $p = 2$ as an exercise on the last sheet, and focus on the case of $p > 2$ starting now. If $p > 2$, then

$\mathbb{Q}(\{2,p\},2) = \{\pm 1, \pm 2, \pm p, \pm 2p\}$ has size eight, so it seems a priori that to use 1.21.2 we need to study eight cases, but the situation is actualy simpler than that.

We know that $[b'] = [-p] \in \operatorname{im}\alpha'$ automatically. The only subgroups of $\mathbb{Q}(\{2,p\},2) = \{\pm 1, \pm 2, \pm p, \pm 2p\}$ that contain $\{[1],[-p]\}$ are

$$\{[1],[-p]\}, \quad \{[\pm 1],[\pm p]\}, \quad \{[1],[2],[-p],[-2p]\}, \quad \{[1],[-2],[-p],[-2p]\},$$

and everything. Therefore, to determine the image of α', it suffices to study three cases: namely, the cases $b_1 = \pm 2$ and $b_1 = p$. By 1.21.2, we need to study whether the following homogeneous spaces have any points:

(a) the case $b_1 = 2$ corresponds to $w^2 = 2u^4 - 2pv^4$,
(b) the case $b_1 = -2$ corresponds to $w^2 = -2u^4 + 2pv^4$, and
(c) the case $b_1 = p$ corresponds to $w^2 = pu^4 - 4v^4$.

These equations are hard to solve in general. Let's look at specific cases; I'll do a few and leave a few more for you to study on the last sheet.

(i) Let's consider $p = 5$. The space corresponding to $b_1 = 2$ has the form $w^2 = 2u^4 - 10v^4$, but it is easy to see that this equation is locally obstructed at 5. Indeed, suppose to the contrary that there is a nontrivial solution (u,v,w), and clear denominators to assume that $u,v,w \in \mathbb{Z}$ with $\gcd(u,v) = 1$. At this point, we note that $5 \nmid u$ because if $5 \mid u$ then $5 \mid w$ and $5 \mid v$ as well, which can't happen. Since $5 \nmid u$, we conclude from $w^2 = 2u^4 - 10v^4$ that 2 is a square mod 5, which is not true. A very similar argument shows that $b_1 = -2$ won't work either. It remains to study the case $b_1 = 5$, which corresponds to the equation $w^2 = 5u^4 - 4v^4$. This equation evidenlty has the nontrivial solution $(1,1,1)$, and so we conclude from this analysis that $\operatorname{im}\alpha' = \{[\pm 1],[\pm 5]\}$. This along with 1.22.4 tells us that $\operatorname{rank} E_5 = \operatorname{rank} E_5' = 1$; in particular, we have shown that $E_5 : y^2 = x^3 + 5x$ has infinitely many solutions. We can easily use these ideas to find generators of $E_5(\mathbb{Q})/2E_5(\mathbb{Q})$, but to go from that to generators of the $E_5(\mathbb{Q})$ needs a little argument involving heights; this I will again leave for the Exercise Sheet.

(ii) More generally, if $p \equiv 5 \pmod 8$, then the same argument as in (i) using that $\left(\frac{\pm 2}{p}\right) = -1$ shows that the values $b_1 = \pm 2$ don't work. This leaves us with the case of $b_1 = p$ corresponding to $w^2 = pu^4 - 4v^4$. Note that by 1.22.4, the equation $w^2 = pu^4 - 4v^4$ has a nontrivial solution iff $\operatorname{rank} E_p = 1$. We can certainly try to solve the above equation for u, v, w–or prove there are no nontrivial solutions–for specific values of p, but doing this for all such p simultaneously seems very hard. We could start by finding local obstructions, but it is not too hard to use Hensel's Lemma to show that this equation is everywhere locally unobstructed, i.e., has solutions in $\mathbb{Q}_q$ for each prime $q \le \infty$. This is sadly not sufficient to conclude that we always have global solutions. Indeed, this is an open problem: it is conjectured, but not yet known, that for all primes p with $p \equiv 5 \pmod 8$, the curve E_p has rank 1.

(iii) Finally, let's consider the case $p = 17$. The case $b_1 = 2$ corresponds to the equation

$$w^2 = 2u^4 - 34v^4. \tag{1.22.15}$$

This equation is very interesting; that something interesting is going on in this example was discovered essentially independently by Carl-Erik Lind, Hans Reichardt, and Gunnar Billing (see this fascinating article [18] by three authors, with two being our very own Paul Pollack and Dan Shapiro discussing the history and mathematics of this specific equation). Suppose this equation has a nontrivial solution (u,v,w), and assume again that $u,v,w \in \mathbb{Z}$ with $\gcd(u,v) = 1$. Note that $w \neq 0$.

Let q be a prime dividing w. If $q = 17$, then $17 \mid u$ too, and then dividing throughout by 17 shows that $17 \mid v$ as well, which can't happen. Therefore, $q \neq 17$. Now if $q \mid u$ or $q \mid v$, then we would conclude from 1.22.15 that q divides both u and v (check!), which is a contradiction to $\gcd(u,v) = 1$. Therefore, $q \nmid uv$, and then we conclude from taking 1.22.15 mod q that $\left(\frac{17}{q}\right) = 1$. Quadratic reciprocity then implies that $\left(\frac{q}{17}\right) = 1$ as well (check!). Since this is true of every prime $q \mid w$ (we do not exclude the cases $w = \pm 1$), and since $17 \equiv 1 \pmod 4$, this is sufficient to conclude that $\left(\frac{w}{17}\right) = 1$, i.e., that w is a quadratic residue mod 17. Finally, consider the equation 1.22.15 mod 17. If $17 \mid u$ or $17 \mid w$, then 17 divides both u and w, and then 1.22.15 implies that $17 \mid v$ as well, which is a contradiction. Since $\left(\frac{w}{17}\right) = 1$, it then follows from 1.22.15 that 2 is a *quartic* residue mod 17, but it is in fact not. This is the contradiction that proves that 1.22.15 has no

rational solution, and hence that $[2] \notin \operatorname{im}\alpha'$. I will leave the analysis of the cases $b_1 = -2$ and $b_1 = 17$ in this case to you as an exercise on the last sheet.
Let us note something very interesting going on here. It can be shown by a very easy application of Hensel's Lemma that 1.22.15 is solvable locally over $\mathbb{Q}_q$ for each prime $q \leq \infty$. Therefore, the fact that it is not solvable over $\mathbb{Q}$ is truly a global phenomenon, and indeed a counterexample to the local-to-global principle for such objects. In the language specific to elliptic curves, this principal homogeneous space (1.22.15) corresponds to a nontrivial element of the Tate-Shafarevich group $Ш(E_{17})$ of the curve $E_{17}/\mathbb{Q}$ (see [8, §X.4]).

Another counterexample of the local-to-global principle (Selmer's counterexample, 1.2.7) will be left for you to analyze on the last sheet.

Remark 1.22.16. The above example (especially the case of the general p with $p \equiv 5 \pmod 8$ and the case of $p = 17$) shows that in general it is not easy to determine which homogeneous spaces arising in the above computation have rational points, since these spaces don't follow the local-to-global principle. This is precisely what makes all current proofs of Mordell-Weil nonconstructive; we just have no algorithm guaranteed to find points on homogeneous spaces or prove that they don't exist.

1.22.C Mazur's Torsion Theorem and Rank Records

We have seen now that we can attach two important arithmetic invariants to an elliptic curve $E/\mathbb{Q}$, namely its rank $r := \operatorname{rank} E$, and its torsion subgroup $T = \operatorname{Tors} E(\mathbb{Q})$. It is a very natural question to ask what values of r and what isomorphism types of T are possible for elliptic curves over $\mathbb{Q}$.

The case of isomorphism types of torsion subgroups is well-understood; the definitive result along this direction is

Theorem 1.22.17 (Mazur)**.** Let $E/\mathbb{Q}$ be an elliptic curve. Then $\operatorname{Tors} E(\mathbb{Q})$ is isomorphic to one of the following 15 groups:

(a) $\mathbb{Z}/n\mathbb{Z}$ for $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$, or 12, or
(b) $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ for $n = 2, 4, 6, 8$.

Each of these groups appears as the torsion subgroup of infinitely many elliptic curves $E/\mathbb{Q}$.

Remark 1.22.18.

(a) This theorem (1.22.17)–conjectured first by Italian mathematician Beppo Levi (1875-1961) in 1908, but known more colloquially as Ogg's conjecture after American mathematician Andrew Ogg (b. 1931)–was proven by the American mathematician Barry Mazur (b. 1937) in 1978, although several special cases (such as the nonexistence of eleven- or sixteen-torsion) were known previously by the work of Levi, Lind Billing-Mahler, and others, which used only elementary techniques.
(b) For each of the isomorphism types allowed by Mazur's Theorem, one can write down explicit families of elliptic curves with that specific torsion subgroup; one such table can be found in [13, Appendix E]. We will discuss some of these "torsion families" when we meet the Tate normal form for elliptic curves next week.
(c) It is a very natural question to ask what the analog of Mazur's Theorem is over arbitrary number fields. A detailed account of this story can be found on the Wikipedia page about Mazur's Theorem, to which I direct the interested reader.

The question of allowable ranks is much more subtle. As of this writing (2026), it is an open problem whether infinitely many values of r are possible, or equivalently if the set of allowable r is bounded. We have evidence that suggests that this should not be the case (e.g., the result is false in the "geometric case", i.e., if we replace $\mathbb{Q}$ by a function field such that $\mathbb{F}_p(t)$), but this is not proof. The curve with the current rank record (as of 2026) of $r \geq 29$ was found by Noam Elkies and Zev Klagsbrun in 2024 and has the form

$$E : y^2 + xy = x^3 + a_4 x + a_6,$$

where

$$a_4 = -27006183241630922218434652145297453784768054621836357954737385$$

and

$$a_6 = 5525805855134237647573669959111819182152106703253507960837240477914941327771617342563672149 7.$$

The rank of this curve is expected to be precisely 29 (this follows from the generalized Riemann Hypothesis), but this fact is not yet unconditionally known. For much more on the history of rank records, rank records with specified torsion, and various other things of this sort, I direct the reader to the website maintained by Croatian mathematician Andrej Dujella, linked here.